BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 153 · Lesson 153 · Integrated Web Forms: HTML, CSS, JavaScript and Server Checks

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Which property writes safe plain text in DOM?

Q2. What is a database foreign key?

Q3. What is the role of a router?

Q4. What does a cache hit mean?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

client-side — Execution on user device.

server-side — Execution on controlled backend.

validation — Checking input against rules.

trust boundary — Point where untrusted data enters a protected system.

New learning

Capture — HTML labels and input types structure a form.

Respond — CSS and JS can give immediate, accessible feedback.

Trust the server — Validate submission and permissions on trusted backend.

Worked example

An online form permits Years 12 or 13. Where must that rule be checked?

1. HTML accepts input — Identify the input or diagram and predict the first result.

2. Browser checks locally — Follow the second stage and explain its importance.

3. Server revalidates — HTML min/max and JavaScript improve normal experience; the server must check the supplied value independently. Client code can be changed or bypassed; no database update should trust it.

We Do · Practise with guidance

A pupil changes a visible price field before submitting checkout. How should the server protect it?

Client total altered

Identify the input or diagram and predict the first result.

Server calculates price

Follow the second stage and explain its importance.

Authorise real total

Calculate actual price using a server-controlled catalogue, validate quantity and permissions, and use authoritative stock records; never trust a submitted total.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Distinguish roles of HTML, CSS and JavaScript in a form. [6]

Question 2

Explain why HTML required alone does not protect database data quality. [6]

Question 3

Design an accessible 16–19 age registration flow with client and server validation. [8]

Exit Ticket · Answer before checking

A number input with min/max means a malicious client cannot submit other values.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. textContent.

Q2. A reference to a key in another table.

Q3. Forward packets between IP networks.

Q4. Required data already present in cache.

You Do pages · Independent questions

Q1. Content and structure [2], presentation [2], client interaction [2].

Q2. Client bypassable [2], crafted requests possible [2], server validation essential [2].

Q3. Associated label and numeric control [2], client feedback [2], server rechecks type/range [2], accessible error and safe storage [2].

Exit Ticket · Model answer

False. Server must revalidate.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: